考试科目名称 计算机网络（期中卷）

2025——2026 学年第 一 学期　　教师 张渊　　考试方式：闭卷

系（专业）＿＿＿＿＿＿＿＿＿＿　　年级＿＿＿＿　　班级＿＿＿＿

学号＿＿＿＿＿＿＿＿　　姓名＿＿＿＿＿＿　　成绩＿＿＿＿

题号	一	二	三	四	五	六	七	八	合计
分数									

得分 ＿＿　一、简答题（本题满分 10 分）

TCP/IP 协议栈是支撑 Internet 的一系列网络协议的总和，请列举出其包含的五个结构层，并简述每层的主要功能。

Score ＿＿　1. Short Answer Questions (Full Scores: 10 points)

The TCP/IP protocol suite is a collection of network protocols that support the Internet. Please list its five structural layers and briefly describe the main functions of each layer.

得分 ＿＿　二、简答题（本题满分 5 分）

持续性 HTTP 和非持续性 HTTP 的区别是什么？

Score ＿＿　2. Short Answer Question (Full Scores: 5 points)

What is the difference between persistent HTTP and non-persistent HTTP?

得分	

三、简答题（本题满分 10 分）

请简述客户端和服务器使用 TCP 协议建立连接的“三次握手”的主要流程，以及 TCP 协议要进行三次握手而不是二次握手的主要原因。

Score	

3. Short Answer Question (Full Scores: 10 points)

Please briefly describe the main process of the "three-way handshake" used by clients and servers to establish a TCP connection, as well as the main reason why the TCP protocol requires a three-way handshake instead of a two-way handshake.

得分	

四、简答题（本题满分 10 分）

请描述 TCP Reno 协议在碰到“3 个重复 ACK 消息”事件和“超时”事件时分别如何设置慢启动 ssthresh 阈值，以及随后拥塞窗口值 CWND 如何增加？

Score	

4. Short Answer Question (Full Scores: 10 points)

Please describe how the TCP Reno protocol sets the slow start threshold (ssthresh) upon encountering a "triple duplicate ACK" event and a "timeout" event, respectively, and how the congestion window (CWND) is increased afterward.

得分	

五、简答题（本题满分 10 分）

dig 是一个向 DNS 服务器发送查询 DNS 记录的工具。图 1 是使用 dig 工具查询南京大学电子邮箱系统域名 A 类记录时的查询请求部分(QUESTION SECTION)和查询结果部分(ANSWER SECTION)的截图。

a) 请问 A 类记录是该域名对应服务器的什么信息？(2 分)
b) 图 1 中 CNAME 记录是由南京大学的服务器返回还是由腾讯 qq 的服务器返回？（2 分）
c) 图 2 是常见 DNS 查询的流程示意图。其中箭头上的数字表示查询请求或答案的发生顺序，请指出每个序号代表的是何种查询方式（迭代或递归）？（4 分）
d) 不考虑 DNS 记录缓存，假设本地 DNS 服务器到所有根服务器、顶级域服务器、权威服务器的 RTT 与图 2 中一致，分别为 RTT_2, RTT_3 和 RTT_4，那么完成 a)中的查询需要耗时多少？（2 分）

Score	

5. Short Answer Question (Full Scores: 10 points)

Dig is a tool used to send queries for DNS records to DNS servers. Figure 1 shows a screenshot of the query request section (QUESTION SECTION) and the query result section (ANSWER SECTION) when using the dig tool to query the A record for the domain name of Nanjing University's email system.

a) What information about the server corresponding to the domain name does the A record provide? (2 points)
b) Does the CNAME record in figure 1 come from Nanjing University's server or from Tencent QQ's server? (2 points)
c) Figure 2 is a schematic of a common DNS query process. The numbers on the arrows indicate the sequence of query requests or responses. Please identify what type of query (iterative or recursive) each number represents. (4 points)
d) Ignoring DNS record caching and assuming the round-trip time (RTT) from the local DNS server to all root servers, top-level domain (TLD) servers, and authoritative servers is consistent with the values shown in figure 2—specifically RTT_2, RTT_3, and RTT_4—how much time would be required to complete the query described in a)? (2 points)

```
;; QUESTION SECTION:
;mail.nju.edu.cn.               IN      A

;; ANSWER SECTION:
mail.nju.edu.cn.        0       IN      CNAME   ssl.exmail.qq.com.
ssl.exmail.qq.com.      0       IN      A       14.17.27.155
```

图 1 使用 dig 返回的结果

Figure 1 results by the dig tool

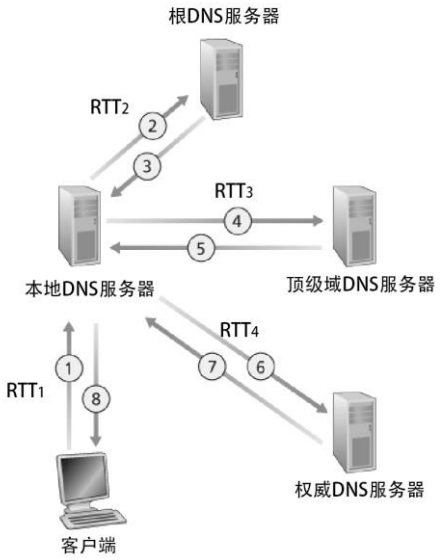


图 2 DNS 查询的流程示意图

Figure 2 DNS query diagram

得分	

六、计算题（本题满分 20 分）

假设你正在从位于北京的实验室计算机（主机 A）向位于上海的远程数据中心服务器（主机 B）传输一个大小为 8 Mbytes（即 8×10^6 bytes）的大文件。数据需要经过以下路径：1）首先通过一条 1 Gbps 的局域网链路连接到北京的一台路由器（R1）；2）然后通过一条 100 Mbps 的长途骨干网链路，中途经由一台位于南京的路由器（R2）进行转发，最终到达上海的数据中心路由器（R3）；3）最后通过另一条 1 Gbps 的局域网链路送达主机 B。
已知物理链路的信号传播速度均为 2×10^8 m/s。北京到南京的物理距离约为 900 km，南京到上海的物理距离约为 250 km。假设所有路由器的处理延时（节点处理延时）均为 1 ms，并且在进行该文件传输时，网络中没有其他背景流量。

a) 请计算从主机 A 发送出该文件的第一个比特开始，到主机 B 接收到该文件的最后一个比特为止，所经历的总时间。（10 分）
b) 如果文件被分为两个相同大小的数据包依次发送，请再次计算经历的总时间。假设路由器必须发送完上一个数据包才开始处理下一个数据包。（10 分）

Score	

6. Calculation Question (Full Scores: 20 points)

Assume you are transferring a large file of size 8 Mbytes (i.e., 8×10^6 bytes) from a laboratory computer (Host A) in Beijing to a remote data center server (Host B) in Shanghai. The data traverses the following path: 1) First, it passes through a 1 Gbps local area network link to a router in Beijing; 2) It is then forwarded via a 100 Mbps long-distance backbone link, passing through a router in Nanjing, ultimately reaching the data center router in Shanghai; 3) Finally, it is delivered to Host B through another 1 Gbps local area network link.
The signal propagation speed on all physical links is 2×10^8 m/s. The physical distance from Beijing to Nanjing is approximately 900 km, and from Nanjing to Shanghai is approximately 250 km. Assume the processing delay at all routers is 1 ms, and there is no other background traffic during this file transfer.

a) Please calculate the total time elapsed from when Host A sends the first bit of the file until Host B receives the last bit of the file. (10 points)
b) If the file is divided into two data packets of equal size and sent sequentially, please calculate the total time elapsed again. Assume routers start to process a packet only after it finishes transmitting the packet that is ahead of this one. (10 points)

得分	

七、计算题（本题满分 15 分）

使用 TCP 协议持续发送数据时，会发现吞吐率(Throughput)会持续出现如图 3 类似的锯齿状波动。记 TCP 发送窗口大小为 WND（单位为 MSS/RTT，其中 MSS 为 Maximum Segment Size，即 TCP 报文长度，RTT 为 Round-trip Time，即传输往返时长）。

a) 请简要解释出现锯齿波动的原因。（5 分）

b) 忽略 TCP 接收端对发送窗口大小的影响，假设网络持续发送数据，并达到一个稳定的波动状态(此时发送窗口大小稳定在 W/2 和 W 之间波动)，记此时网络的丢包率(Loss Rate)大小为 L，请给出一个使用 W 来估算 L 的表达式。（5 分）

c) 对于较大的 W 值，请证明稳定状态下的 TCP 平均吞吐率可用以下公式估算：

$$1.22\frac{MSS}{RTT \times \sqrt{L}}$$

已知 $\sqrt{1.5} \approx 1.22$。（5 分）

Score	

7. Calculation Question (Full Scores: 15 points)

When using the TCP protocol to continuously send data, it can be observed that the throughput exhibits a sawtooth-like fluctuation pattern as shown in the figure below. Let the TCP sending window size be denoted as WND (in units of MSS/RTT, where MSS is the Maximum Segment Size, i.e., the length of a TCP segment, and RTT is the Round-trip Time, i.e., the transmission round-trip duration).

a) Please briefly explain the reason for the sawtooth fluctuation. (5 points)

b) Ignoring the impact of the TCP receiver on the sending window size, assume the network continuously sends data and reaches a stable fluctuating state (where the sending window size fluctuates steadily between W/2 and W). Denote the packet loss rate of the network at this point as L. Please provide an expression to estimate L using W. (5 points)

c) For a large value of W, please prove that the average TCP throughput in the steady state can be estimated using the following formula:

$$1.22\frac{MSS}{RTT \times \sqrt{L}}$$

Given that $\sqrt{1.5} \approx 1.22$. (5 points)

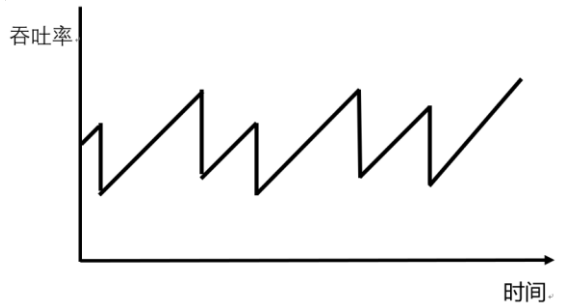


图 3 吞吐率呈现的锯齿状波动

Figure 3 The throughput exhibits sawtooth-like fluctuations

得分	

八、设计题（本题满分 20 分）

考虑使用 P2P 架构将大小为 F 比特的文件分发给 N 个对等方。假设采用流体模型，即只要速率之和不超过 u_S，服务器就可以同时向多个对等方传输，并以不同速率向每个对等方传输。为简化分析，假定 d_{min}（最小下载带宽）极大，因此对等方的下载带宽永远不会成为瓶颈。

a) 假设 $u_S \leq (u_S + u_1 + u_2 + \cdots + u_N)/N$，请你设计一个分发方案，使得完成分发的总时间为 F/u_S。（10 分）

b) 假设 $u_S \geq (u_S + u_1 + u_2 + \cdots + u_N)/N$，请你设计一个分发方案，使得完成分发的总时间为 $NF/(u_S + u_1 + u_2 + \cdots + u_N)$。（10 分）

Score	

8. Design Question (Full Scores: 20 points)

Consider distributing a file of F bits to N peers using a P2P architecture. Assume a fluid model where the server can simultaneously transmit to multiple peers, transmitting to each peer at different rates, as long as the combined rate does not exceed u_S. For simplicity assume that minimum download rate d_{min} is very large so that peer download bandwidth is never a bottleneck.

a) Suppose that $u_S \leq (u_S + u_1 + u_2 + \cdots + u_N)/N$. Specify a distribution scheme that has a distribution time of F/u_S. (10 points)

b) Suppose that $u_S \geq (u_S + u_1 + u_2 + \cdots + u_N)/N$. Specify a distribution scheme that has a distribution time of $NF/(u_S + u_1 + u_2 + \cdots + u_N)$. (10 points)